

Smart Home Automation System using BLYNK

Project Aim

Design and implement a Smart Home Automation System using ESP32 and Blynk for automatic light and fan control with environmental monitoring.

Project Overview

This project uses an ESP32 with Blynk IoT to automate a smart home. An LDR sensor controls the room light based on brightness, while a DHT22 sensor monitors temperature and humidity. A fan operates automatically when temperature exceeds 30°C. All parameters and device statuses are displayed on the Blynk mobile dashboard.

Components Used

- ESP32 DevKit V1
- DHT22 Temperature and Humidity Sensor
- LDR Sensor
- 10k Ω Resistor (for voltage divider with LDR)
- LED (Room Light)
- LED (Fan Indicator)
- Buzzer
- 220 Ω Resistors $\times$ 2
- Blynk

Construction and Wiring Details

Component	ESP32 Pin
DHT22 Data	GPIO 15
LDR Output	GPIO 34
Room Light LED (+)	GPIO 18
Fan LED (+)	GPIO 19
Buzzer (+)	GPIO 23
DHT22 VCC	3.3V
DHT22 GND	GND
LDR One Side	3.3V
LDR Other Side	GPIO34 and 10kΩ to GND
LEDs Cathode	GND
Buzzer (-)	GND

Blynk Connection

Datastream	Virtual Pin	Type
Temperature	V0	Double
Humidity	V1	Double
Light Status	V2	String
Fan Status	V3	String
Brightness Value	V4	Integer

Code Structure / Program Flow

Start



Initialize ESP32



Connect to WiFi & Blynk



Read DHT22 Temperature
and Humidity Values

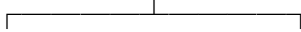


Read LDR Value

|



Is LDR value < 2000?



Yes

No

|

|



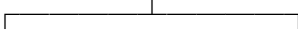
Light ON

Light OFF

|



Is Temperature > 30°C ?



Yes

No

|

|



Fan ON

Fan OFF

Buzzer ON

Buzzer OFF

|



Send Data to Blynk

|



Repeat Continuously

Source Code

Diagram.jetson

```
{
  "version": 1,
  "author": "Aman Abdulla",
  "editor": "wokwi",
  "parts": [
    { "type": "board-esp32-devkit-c-v4", "id": "esp", "top": 19.2, "left": 110.44, "attrs": {} },
    { "type": "wokwi-dht22", "id": "dht1", "top": -95.7, "left": 311.4, "attrs": {} },
    {
      "type": "wokwi-photoresistor-sensor",
      "id": "ldr1",
      "top": -16,
      "left": -181.6,
      "attrs": {}
    },
    {
      "type": "wokwi-buzzer",
      "id": "bz1",
      "top": 69.6,
      "left": -103.8,
      "attrs": { "volume": "0.1" }
    },
    { "type": "wokwi-led", "id": "led1", "top": 54, "left": 397.4, "attrs": { "color": "red" } },
    { "type": "wokwi-led", "id": "led2", "top": 54, "left": 464.6, "attrs": { "color": "green" } },
```

```

{
  "type": "wokwi-resistor",
  "id": "r1",
  "top": 129.6,
  "left": 393.05,
  "rotate": 90,
  "attrs": { "value": "220" }
},
{
  "type": "wokwi-resistor",
  "id": "r2",
  "top": 129.6,
  "left": 460.25,
  "rotate": 90,
  "attrs": { "value": "220" }
}
],
"connections": [
  [ "esp:TX", "$serialMonitor:RX", "", [] ],
  [ "esp:RX", "$serialMonitor:TX", "", [] ],
  [ "dht1:GND", "esp:GND.2", "black", [ "v28.8" ] ],
  [ "esp:15", "dht1:SDA", "gold", [ "v0", "h115.2" ] ],
  [ "esp:3V3", "dht1:VCC", "magenta", [ "h-9.45", "v-38.4", "h172.8", "v48", "h48" ] ],
  [ "ldr1:VCC", "esp:3V3", "red", [ "h76.8", "v57.6" ] ],
  [ "ldr1:GND", "esp:GND.1", "black", [ "h115.2", "v162.8" ] ],
  [ "esp:34", "ldr1:AO", "green", [ "h-57.45", "v-19.2" ] ],

```

```

[ "esp:GND.1", "bz1:1", "black", [ "h-191.85", "v9.6" ] ],
[ "esp:23", "bz1:2", "green", [ "h28.8", "v182.4", "h-297.6" ] ],
[ "led1:A", "r1:1", "green", [ "v0" ] ],
[ "led2:A", "r2:1", "green", [ "v0" ] ],
[ "esp:18", "r1:2", "green", [ "h144", "v76.8", "h9.6" ] ],
[ "esp:19", "r2:2", "blue", [ "h163.2", "v134.4", "h9.6" ] ],
[ "esp:GND.2", "led1:C", "black", [ "v0", "h172.8", "v57.6" ] ],
[ "esp:GND.2", "led2:C", "black", [ "v0", "h249.6", "v48" ] ]
],
"dependencies": {}
}

```

Esp32 Code

```

#define BLYNK_TEMPLATE_ID "TMPL3XnEF25p"

#define BLYNK_TEMPLATE_NAME "Smart Home Automation System"

#define BLYNK_AUTH_TOKEN "sEZ1iUVVjmSO7xLNCGVHGyI5h3rVP5_m"


#include <WiFi.h>

#include <BlynkSimpleEsp32.h>

#include <DHTesp.h>


char ssid[] = "Wokwi-GUEST";

char pass[] = "";

```

```
#define DHTPIN 15  
  
#define LDR_PIN 34  
  
#define LIGHT_LED 18  
  
#define FAN_LED 19  
  
#define BUZZER 23
```

```
DHTesp dht;
```

```
BlynkTimer timer;
```

```
void sendSensorData()  
{  
  TempAndHumidity data = dht.getTempAndHumidity();
```

```
  float temperature = data.temperature;
```

```
  float humidity = data.humidity;
```

```
  int ldrValue = analogRead(LDR_PIN);
```

```
  // Send values to Blynk
```

```
  Blynk.virtualWrite(V0, temperature);
```

```
  Blynk.virtualWrite(V1, humidity);
```

```
  Blynk.virtualWrite(V4, ldrValue);
```

```
  // Light control
```

```
  if (ldrValue < 2000)
```



```
{  
    digitalWrite(LIGHT_LED, HIGH);  
    Blynk.virtualWrite(V2, "Light ON");  
}  
else  
{  
    digitalWrite(LIGHT_LED, LOW);  
    Blynk.virtualWrite(V2, "Light OFF");  
}
```

```
// Fan control
```

```
if (temperature > 30)  
{  
    digitalWrite(FAN_LED, HIGH);  
    digitalWrite(BUZZER, HIGH);  
    Blynk.virtualWrite(V3, "Fan ON");  
}  
else  
{  
    digitalWrite(FAN_LED, LOW);  
    digitalWrite(BUZZER, LOW);  
    Blynk.virtualWrite(V3, "Fan OFF");  
}
```

```
Serial.print("Temperature: ");
```

```
Serial.print(temperature);
```

```
    Serial.print(" °C Humidity: ");  
    Serial.print(humidity);  
    Serial.print("% LDR: ");  
    Serial.println(ldrValue);  
}
```

```
void setup()
```

```
{
```

```
    Serial.begin(115200);
```

```
    pinMode(LIGHT_LED, OUTPUT);
```

```
    pinMode(FAN_LED, OUTPUT);
```

```
    pinMode(BUZZER, OUTPUT);
```

```
    dht.setup(DHTPIN, DHTesp::DHT22);
```

```
    Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);
```

```
    timer.setInterval(2000L, sendSensorData);
```

```
}
```

```
void loop()
```

```
{
```

```
    Blynk.run();
```

```
    timer.run();
```

```
}
```

Project Photograph



Smart Home Automation System

Dashboard

📌 This is how the device page will look like for actual devices.

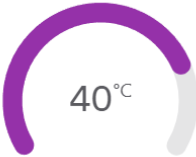


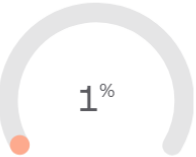
Device Name ● Online

[👤 Device Owner](#) [🏢 Company Name](#)

🔔 🔧 📄 ⌵ ⋮

1h6h1d1w1mo3mo📅

temperature (V0)

050

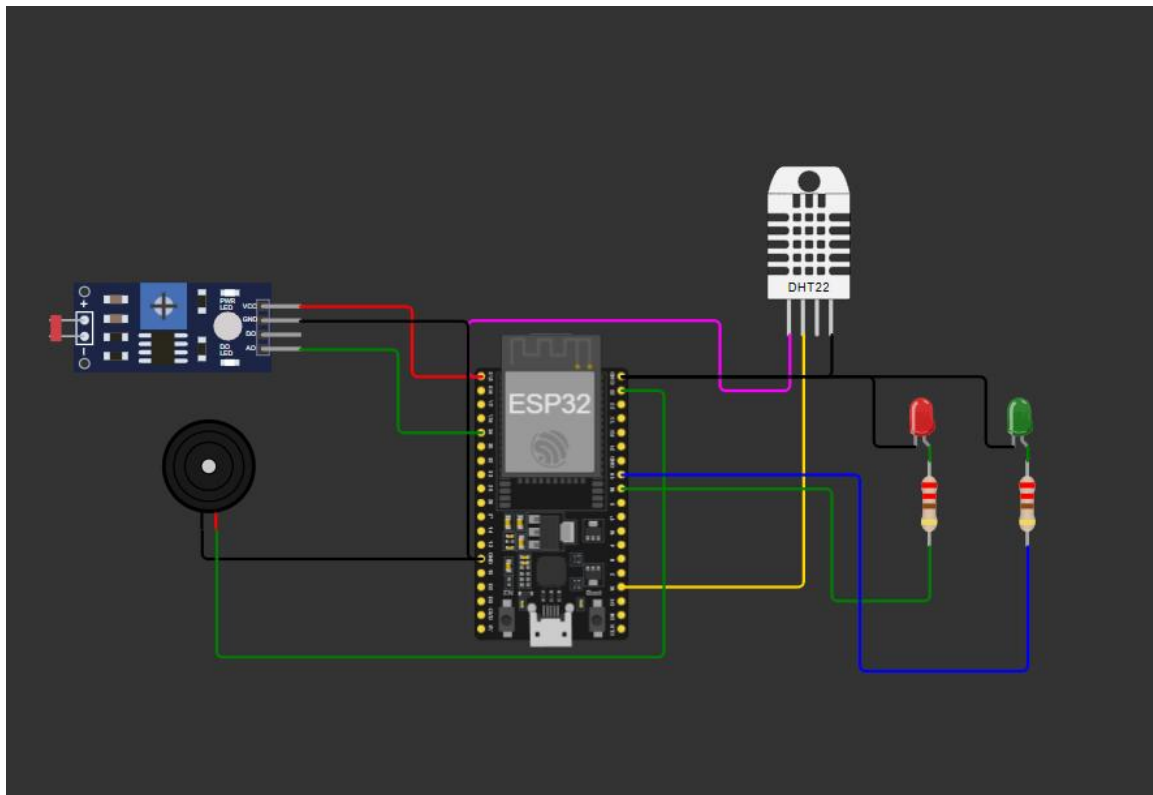
humidity (V1)

0100

Brightness Value L... (V4)

04,095

light (V2)
String

fan (V3)
String



Links

Google drive :

https://drive.google.com/drive/folders/1_8HS2EsHL1lLBestaE9jH79jfWASTdkm?usp=sharing

Wokwi: <https://wokwi.com/projects/466601254837816321>

Git hub repo : <https://github.com/AmanAZ5950/CIRCUITRON.git>